Commonlands LLC
2640 Financial Ct Suite A
San Diego CA 92117
www.commonlands.com
+1 (858) 333-7325

CIL212 Product Specification

Find out more at: commonlands.com/cil212

Table of Contents

Table of Contents 2
Revision History 3
Overview 4
Description 4
Features 4
Applications 4
Optical Specifications 5
First Order Specifications 5
Field of View Table 5
Mechanical Specifications: 6
Camera Mechanical Compatibility Table 6
CIL212-F2.2-M12A660 (with IR Cut-Off Filter) 6
CIL212-F2.2-M12ANIR (without IR Cut-Off Filter) 7
Compatible Accessories 8
Additional Information 9
Related Files 9
Contact Us 9
Disclaimer 9

Revision History

Revision	Date	Description
1.0	July 8, 2022	Initial Release

Overview

Description
226°@3.9mm Fisheye M12 Lens

The CIL212 is a 4K+ 226° M12 fisheye lens with a 3.9mm image circle. The fast F/2.0 aperture makes this lens great for 360° surveillance and computer vision.

Features
- Miniature Size
  - The 20mm TTL is great for doorbell cameras and aerial robotics cameras.
- 226° FoV
  - Create a compact camera with an amazingly large fisheye field of view.

Applications
This lens can be used for many applications. The specifications are well suited for the requirements commonly seen in the following applications:
- Consumer + AR/VR
- Smart Everywhere

Optical Specifications

First Order Specifications

Parameter	Value
Part Number	CIL212
Format Type	1/2.5”
Focal Length (EFL)	1.1mm
F#	2.2 Fixed
Resolution	8-12MP
Maximum Image Circle	3.9mm
Weight	6.0g
IR Cut-Off Filter	P/N With: CIL212-F2.2-M12A660 P/N Without: CIL212-F2.2-M12ANIR
Optical Construction	Hybrid Glass/Plastic
Barrel Material	Aluminum

Field of View Table

Image Circle	Field of View	Distortion		
		Rectilinear	F-Θ	TV (4:3)
1.8mm	96°	-26%	-1.8%	-12%
2.4mm	129°	-48%	-3.4%	-17%
3.9mm	226°	-175%	-10%	-29%

Calculate field of view for your sensor at: commonlands.com/m12-lenses
Calculate distortion at: commonlands.com/pages/fov-calculator

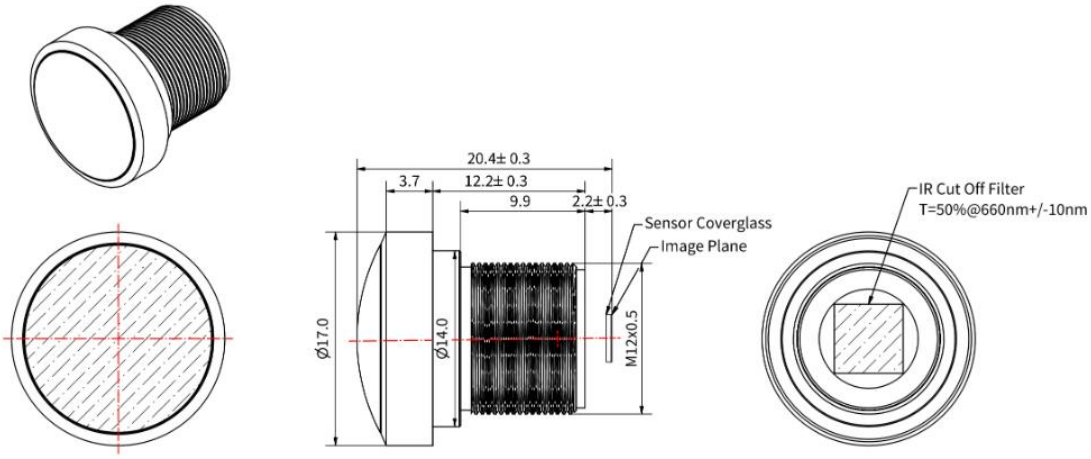
Mechanical Specifications:

Camera Mechanical Compatibility Table

Camera/Mount	Compatibility
e-con Systems Standard S-Mount	With CLA020
Allied Vision Alvium S-Mount	Yes
FLIR Blackfly S Board	No
iDS uEye LE	No
The Imaging Source M12	No
Basler Dart S-Mount	No

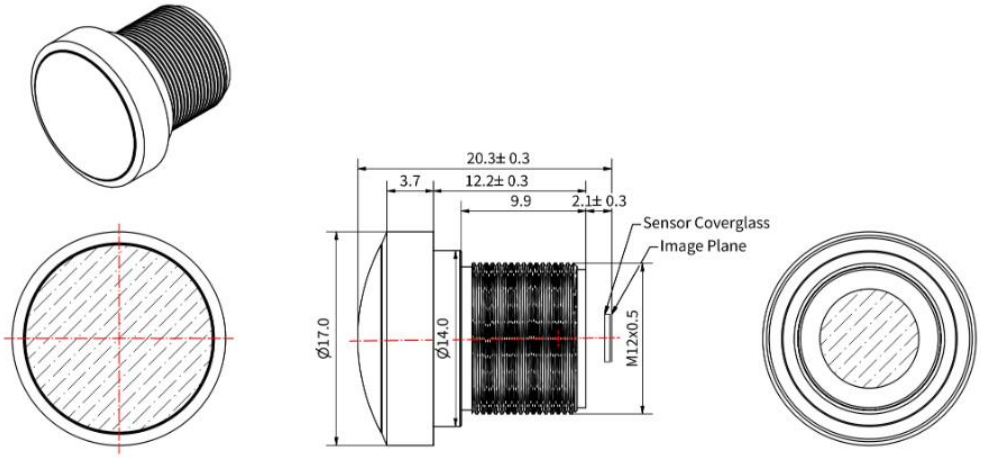
This table is for mechanical compatibility only, assuming the standard M12x0.5 lens mount provided by the camera manufacturer. Please check optical compatibility separately.

CIL212-F2.2-M12A660 (with IR Cut-Off Filter)



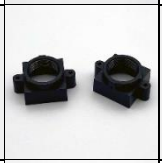
The full release mechanical drawing is available for download at:
Commonlands.com/drawings-cil212-m12a660
The .STP file is available for download at:
Commonlands.com/stp-cil212-m12a660

CIL212-F2.2-M12ANIR (without IR Cut-Off Filter)



The full release mechanical drawing is available for download at:
Commonlands.com/drawings-cil212-m12anir
The .STP file is available for download at:
Commonlands.com/stp-cil212-m12anir

Compatible Accessories

Accessory	Image	Description	Link
CLA018		M12 Lens Mount, 18mm Spacing and 8.5mm Height	Commonlands.com/cla018
CLA020		M12 Lens Mount, 20mm Spacing and 10mm Height	Commonlands.com/cla020
CLA022		M12 Lens Mount, 22mm Spacing and 7.9mm Height	Commonlands.com/cla022
CLA-M12-CS-Adapter		M12 to CS Mount Adapter (w/ Locking Ring)	Commonlands.com/cla-m12-cs-adapter
CLA-M12-C-Adapter		M12 Lens to C CS Mount Adapter - Thin	Commonlands.com/cla-m12-c-adapter
CLA-M12-LockRing-AL6061		M12 Lens to C CS Mount Adapter - Thin	Commonlands.com/cla-m12-lockring-al6061
CLA-M12-LockRing-PC		M12 Lens Focus Lock Nut Ring, PC	Commonlands.com/cla-m12-lockring-pc

A full list of accessories is available at: commonlands.com/collections/accessories

Additional Information

Related Files

Item	Variant	Link
Web Page Quick Link	Both	Commonlands.com/cil212
Download Mechanical Drawing	CIL212-F2.2-M12A660 with IR Filter	Commonlands.com/drawings-cil212-m12a660
Download Mechanical Drawing	CIL212-F2.2-M12ANIR without IR Filter	Commonlands.com/drawings-cil212-m12anir
Download .STP	CIL212-F2.2-M12A660 with IR Filter	Commonlands.com/stp-cil212-m12a660
Download .STP	CIL212-F2.2-M12ANIR without IR Filter	Commonlands.com/stp-cil212-m12anir

Contact Us

Please contact our engineering team if you have any questions about the product specification or would like assistance with product recommendations.

Call us at: +1 (858) 333-7325

Email us at: engineering@commonlands.com

Schedule a call (Zoom or Phone) using: https://calendly.com/commonlands/call

For pricing and availability, please contact sales@commonlands.com. For quantities less than 10, the easiest method to purchase is directly through the online store using the “Web Page Quick Link” URL for this product.

Disclaimer

This document is subject to change without notice. All product purchases, specifications, and warranties are subject to our standard terms and conditions at: commonlands.com/policies/terms-of-service. Any return requests are subject to the refund terms and conditions at: commonlands.com/policies/refund-policy